

The following point-wise notes provide a comprehensive overview of the key topics presented in the sources.

I. Cloud Computing Technology

Clients

- **Mobile Clients:**
 - Include laptops, PDAs, and smartphones.
 - Laptops can access cloud applications similar to desk-bound clients.
 - Security concern: Easier to lose a laptop, risking data compromise.
 - Security benefit: If data is primarily maintained on the cloud, theft of the laptop results in minimal data compromise.
- **Thin Clients:**
 - Computers with no hard drives or DVD-ROM drives; simply display content from the server.
 - Excellent option for accessing cloud-based services or virtualized servers.
 - Advantages: Less expensive, lower maintenance costs, use less energy, and provide a **high level of security** because no data is stored locally.
- **Thick Clients:**
 - Standard computers likely already in use.
 - Suitable if users need to keep local files or run mission-critical applications that must remain in-house.
 - Disadvantages: **More vulnerable to attack** than thin clients, as data stored on the hard drive can be compromised if the machine is stolen. If a thick client fails, data, OS, and configuration settings are lost, requiring a new computer setup.

Security

- Security is the number one issue in cloud computing because data is stored by a third party.
- **Benefits:**
 - **Centralization of data** reduces the likelihood of data leakage often associated with files stored on thick clients and laptops.
 - Cloud providers may offer superior security features due to the economy of scale.
 - Logging can be improved by providers extending memory in the virtualized environment.
- **Response and Forensics:**
 - Providers can respond to breaches with less downtime; a forensic server can be built online easily.
 - A virtual machine can be cloned for easy offline analysis if a problem occurs.
 - Security vendors are actively developing products for virtual machines and the cloud.

- **Compliance and Auditing:**
 - Regulations like SOX, GLBA, HIPAA, and PCI DSS add complexity.
 - It is difficult to determine the physical storage location of compliance-sensitive data (it could be on a provider's partner's equipment).
 - Providers should adhere to **Open Web Application Security Project (OWASP) guidelines** for secure application development.

Key Management and Secure Remote Access

- **Key Management:**
 - Remote services and servers must be **cryptographically authenticated** using client and server certificates.
 - Cloud storage must be cryptographically protected by encrypting data and ensuring data is destroyed when its storage key is destroyed.
 - This process requires managing many keys (transport keys, file encryption keys, authorization tokens, etc.).
- **SSL VPNs:**
 - A **Secure Sockets Layer virtual private network (SSL VPN)** is needed for secure remote access.
 - SSL VPNs can be used with a standard web browser and do not require specialized client software installation (unlike traditional IPsec VPNs).
 - They enforce **endpoint security measures** by verifying running antivirus software, installed OS patches, and checking for malware.

Connectivity Options

1. **Basic Public Internet:**
 - Advantages: Cost-effective, highly fault tolerant, and provides confidentiality via SSL-based encrypted access (HTTPS).
 - Disadvantages: Lack of end-to-end Quality of Service (QoS), making Service-Level Agreements (SLAs) difficult, and poor response time over high-latency connections.
2. **Accelerated Internet:**
 - Improves cloud performance by 20–50% by offloading network-related functions (like SSL termination and TCP connection management) from servers.
 - Dynamic caching, compression, and prefetching can result in over 50% performance increase.
3. **Optimized Internet Overlay:**
 - Access is via the public Internet, but enhancements occur at provider Points of Presence (POP).
 - Enhancements include optimized real-time routing, SSL session optimization, and placing some application logic on the POP.
 - Offers better scalability and response time (usually >80%) but is costlier than public Internet.

4. **Site-to-Site VPN:**

- Connects directly via a private Wide Area Network (WAN), typically MPLS/VPN.
- Provides confidentiality, **guaranteed bandwidth**, and SLAs for latency and packet loss.

Bandwidth and Redundancy

- **Bandwidth:** Refers to transmission speed/throughput.
 - Transfers can be **symmetric** (sending and receiving at the same rate) or **asymmetric** (different rates, e.g., ADSL).
 - SLAs are crucial to guarantee bandwidth requirements and ensure remediation if the ISP fails to meet levels.
- **Redundancy:**
 - Cloud infrastructure must be configured for reliability and failover.
 - In the cloud, redundancy involves reallocating resources, as data may be scattered across various geographic locations within the provider's cloud, rather than wheeling in a new server.

Services

- **Identity Services:** Applications require a digital identity (set of bytes) to determine user permissions.
 - Cloud providers use proprietary identity services (e.g., Amazon, Google App Engine).
 - **OpenID** is an open, decentralized single sign-on standard that uses a URL for identity verification without relying on a central authority.
- **Integration Services:**
 - Amazon's **Simple Queue Service (SQS)** provides application message exchange via queues, although it does not guarantee in-order delivery, which aids scalability.
 - **BizTalk Services** uses a relay service to enable applications to tunnel through firewalls for communication.
- **Other Services:** Cloud functions include Mapping (e.g., Google Maps), Payments (online payment services like PayPal), and Search (submitting searches to internal databases and the Internet, e.g., Microsoft Live Search).

II. Accessing the Cloud

Web Application Frameworks and Techniques

- **Web Application Frameworks:** Support the development of dynamic websites and services, reducing development overhead via pre-written libraries.

- **AJAX (Asynchronous JavaScript and XML):** Techniques for interactive web applications.
 - Retrieves data asynchronously in the background, minimizing interference with the current page display.
 - Technologies leveraged include XHTML, CSS, DOM, XML/XSLT, and JavaScript (or JSON).
 - Increases efficiency, reduces bandwidth consumption, and makes the application appear faster to the user.
- **Python Django:** An open-source web application framework written in Python.
 - Developed for database-driven websites, utilizing the **DRY (Don't Repeat Yourself)** principle.
 - Includes a development web server, a caching framework, and a dynamically generated administrative CRUD interface.

Cloud Hosting and PaaS

- **Amazon EC2 (Elastic Compute Cloud):** Provides resizable compute capacity. Uses Xen virtualization; virtual machines are called instances. Allows quick scaling and a pay-only-for-capacity-used model.
- **Mosso:** Provides enterprise-grade hosting and storage. Components include Cloud Sites (Windows/Linux apps), Cloud Files (unlimited online storage), and Cloud Servers (instantly deployable).
- **Azure Services Platform (Microsoft):** Combines cloud-based developer capabilities with storage, computation, and networking infrastructure. Key components include Windows Azure, SQL Services, and .NET Services.
- **Force.com (Salesforce.com PaaS):** A Platform-as-a-Service for creating and deploying business applications.
 - It is a multitenant platform featuring an on-demand operating system and the **Apex Code** programming language.
 - **Visualforce** is a framework component that allows users to design application UIs using HTML, AJAX, and Flex.
- **Google Apps Premier Edition:** Includes services like Gmail, Calendar, Talk, and Docs/Spreadsheets. Unique features include 10GB storage, 99.9% uptime SLAs, and APIs for business integration.

APIs (Application Programming Interfaces)

- APIs are programming instructions and standards used for accessing web-based programs. They allow programs to communicate silently in the background, typically managed by web services using XML.
- **Google Data APIs:** REST-style APIs based on AtomPub, used to link applications within and outside the cloud (e.g., Google Calendar, YouTube).

- **Apex Web Services API (Salesforce.com):** Used to access and manage complex data relationships in a single request, similar to database JOIN functionality, which enhances speed and simplicity.

Web Browsers

- **Google Chrome:** Built for the modern web, featuring a combined search/address bar.
 - Tabs operate as separate processes; isolating tabs ensures stability if one crashes.
 - Uses the V8 JavaScript engine for speed.
 - Extends cloud capabilities via **built-in Google Gears** components, such as a database module for local data storage, and a LocalServer module for caching resources.

III. Cloud Storage

Storage Model and Reliability

- **Storage as a Service (SaaS):** A third-party provider rents storage space, ideal for organizations lacking the budget or technical personnel for their own infrastructure. Users pay per gigabyte stored or data transferred.
- **Reliability:** Cloud storage achieves reliability primarily through **redundancy**, storing saved data on multiple machines, often using different power supplies, to ensure data access despite failures.

Security and Cautions

- **Security Measures:** Include **encryption** (requiring a key to decode), **authentication** (name and password), and **authorization practices** (defining user access levels).
- **Concerns:** Storing proprietary data off-site poses a security risk, especially if stored alongside a competitor's data. Encryption and securing data transit with SSL are mandatory.
- **Cautions:** Since cloud storage is immature, a **mixed approach** (using it first for noncritical purposes) is recommended. Organizations must vet vendors for price and reliability, as massive outages (like Amazon S3 in 2008) can prevent data access even if data is not lost. Legal issues regarding copyrighted material must also be addressed.

Key Providers and Technologies

- **Amazon S3 (Simple Storage Service):** Designed to be highly scalable, reliable (**99.99% availability**), fast, inexpensive (using commodity hardware), and simple. It uses standards-based REST and SOAP interfaces. Its design adheres to distributed system principles like decentralization, autonomy, and failure toleration.

- **Nirvanix Storage Delivery Network (SDN):** Turns a server into an infinite capacity Network Attached Storage (NAS) file. CloudNAS provides built-in data disaster recovery, encrypted offsite storage, and automated data replication on up to three geographically dispersed nodes.
- **Google Bigtable Datastore:** Highly scalable database developed for speed and flexibility, capable of spanning thousands of distributed servers. It is a multidimensional sparse map where each cell has a timestamp. Its proprietary nature means developers risk vendor lock-in.
- **MobileMe (Apple):** Subscription-based service delivering **push email, contacts, and calendars** to native applications across devices (iPhone, Mac, PC).
- **Live Mesh (Microsoft):** A "software-plus-services" platform that synchronizes files and applications across a user's devices. It includes the Mesh Operating Environment (MOE) software for synchronization and Live Desktop for online storage.

IV. Cloud Standards

Protocols and Messaging

- **HTTP (Hypertext Transfer Protocol):** The computing mechanism used to transfer data. It is a **stateless protocol**, requiring methods like cookies to maintain user state. HTTP 1.1 allows the connection to remain open to transfer all page components, unlike 1.0 which required separate requests.
- **XMPP (Extensible Messaging and Presence Protocol):** Touted as a major standard for cloud computing. It allows for **two-way communication** and eliminates polling (where clients repeatedly ping a host for updates), which is common in older web services APIs.

Security Standards

- **SSL (Secure Sockets Layer):** Standard technology for establishing an encrypted link between a web server and browser. Requires an SSL certificate and private/public key pairing.
- **OpenID:** An open-source, lightweight solution for authentication, simplifying user experience by eliminating the need for a unique username and password for every website (single sign-on).
- **PCI DSS:** Requirement 2.2.1 regarding one primary function per server can be met in virtualized environments by ensuring systems are segmented and isolated.

Data Standards

- **XML (Extensible Markup Language):** A standard, self-describing way of encoding text and data for interoperable exchange across platforms.
 - Tags define the **structure and content** of the data, separating form from content.

- Ideal for cloud computing because data spread across multiple servers can be made to look unified.
- **JSON (JavaScript Object Notation):** A lightweight data interchange format based on a subset of JavaScript.
 - Often preferred over XML when JavaScript is involved because the object is already usable, reducing the need for extra code and overhead associated with extracting data from DOM objects.

Web Services

- **REST (Representational State Transfer):** An architectural style that uses existing web technology (HTTP, XML) to retrieve information by reading a designated web page containing an XML file.
 - It is easier to use than SOAP.
 - Benefits include better response time, reduced server load (due to caching), and improved server scalability by reducing the need to maintain session state.
- **SOAP (Simple Object Access Protocol):** Allows programs on different operating systems to communicate using HTTP and XML.
 - SOAP encodes an HTTP header and XML file to enable program-to-program calls.
 - A major advantage is that SOAP calls are **more likely to pass through firewalls** than traditional Remote Procedure Calls (RPCs) because they use standard HTTP requests.

Infrastructure Standards (Virtualization)

- Major players collaborated to advance open virtualization standards to accelerate solutions and interoperability.
- VMware contributed its ESX Server source code/interfaces via the **Community Source** program to influence the direction of the standard.
- **OVF (Open Virtualization Format):** A platform-independent, extensible, and open specification for packaging virtual appliances.
 - OVF prevents **vendor lock-in** by allowing customers to select any hypervisor based on preference or functionality.